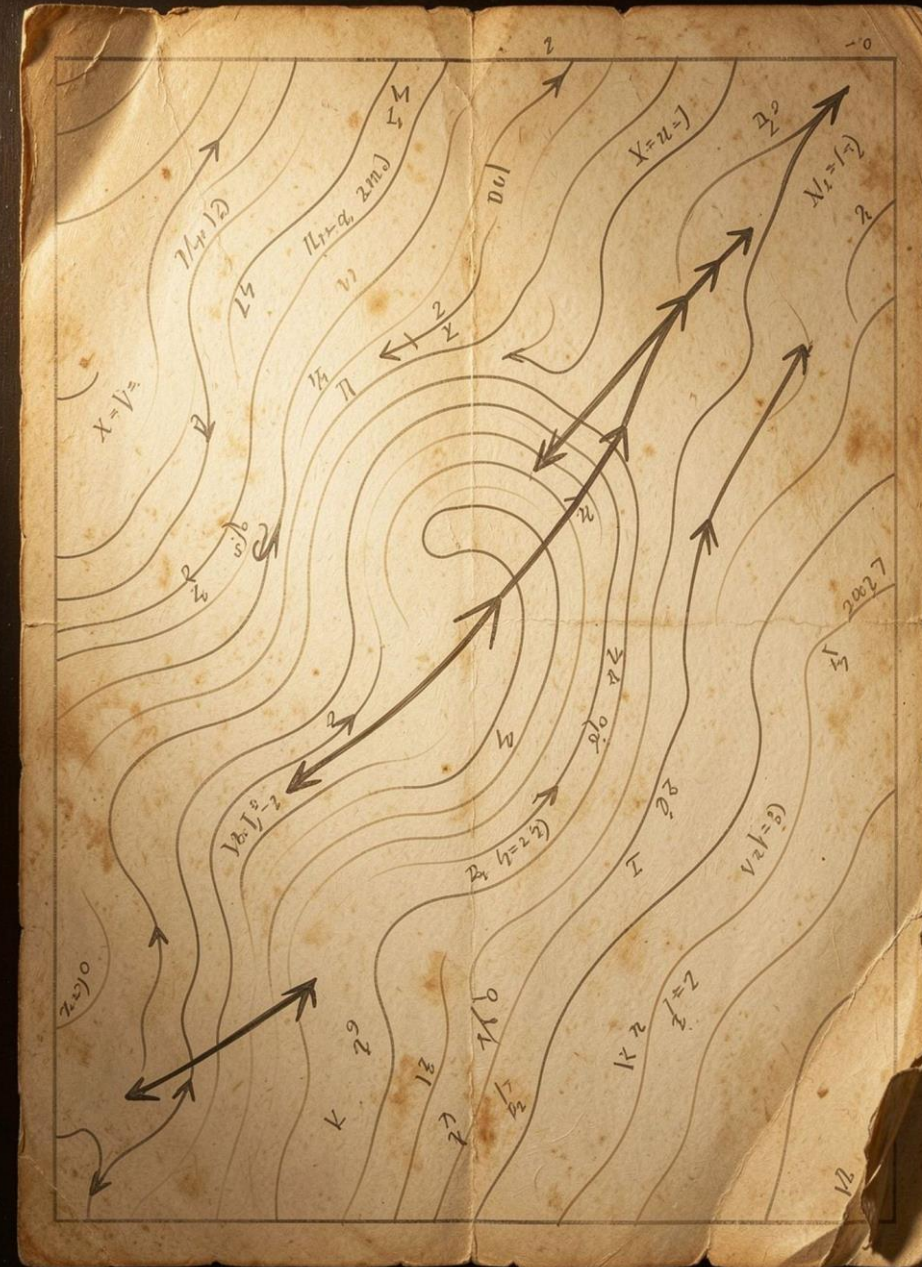


VECTOR ANALYSIS

Gradient and Directional Derivative

A rigorous introduction to the gradient vector and directional derivative — two of the most powerful tools in multivariable calculus. These concepts form the mathematical backbone of heat transfer, fluid dynamics, optimization, and electromagnetics. Designed for BSc Engineering students with a foundation in single-variable calculus and partial differentiation.



Learning Objectives

By the end of this module, you should be able to:

1

Define and Compute the Gradient

Understand the gradient vector as a collection of partial derivatives. Compute it for scalar fields in two and three dimensions, and interpret its geometric meaning as the direction of steepest ascent on a surface.

2

Compute the Directional Derivative

Use the dot product formula to find the rate of change of a function in an arbitrary direction. Understand why the direction vector must be normalized to unit length.

3

Apply to Engineering Problems

Connect these mathematical tools to real engineering contexts: heat flux in thermal systems, velocity potentials in fluid flow, gradient-based optimization, and electric field intensity in electromagnetics.

Question: Suppose, $\vec{P} = xy \hat{i} - yz \hat{j} + z^2 \hat{k}$, $\vec{Q} = y\hat{i} + 2x\hat{j} - 3z\hat{k}$, $\vec{R} = x^2\hat{i} - z^2\hat{j} - y^2\hat{k}$ and $\varphi = xyz$. Show that (i) $(\vec{P} \cdot \vec{\nabla})\varphi = (\vec{\nabla}\varphi) \cdot \vec{P}$ and (ii) $(\vec{R} \cdot \vec{\nabla})\varphi = (\vec{\nabla}\varphi) \cdot \vec{R}$ (*yourself*)

Solution: (i)

$$\text{Here, } \vec{P} \cdot \vec{\nabla} = (xy\hat{i} - yz\hat{j} + z^2\hat{k}) \cdot \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) = xy \frac{\partial}{\partial x} - yz \frac{\partial}{\partial y} + z^2 \frac{\partial}{\partial z}$$

$$\begin{aligned} \text{Therefore } (\vec{P} \cdot \vec{\nabla})\varphi &= \left(xy \frac{\partial}{\partial x} - yz \frac{\partial}{\partial y} + z^2 \frac{\partial}{\partial z} \right) (xyz) \\ &= xy \frac{\partial}{\partial x} (xyz) - yz \frac{\partial}{\partial y} (xyz) + z^2 \frac{\partial}{\partial z} (xyz) \\ &= xy \cdot yz - yz \cdot xz + z^2 \cdot xy = xy^2z \end{aligned}$$

$$\begin{aligned} \text{Again, } \vec{\nabla}\varphi &= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) (xyz) \\ &= \frac{\partial}{\partial x} (xyz) \hat{i} + \frac{\partial}{\partial y} (xyz) \hat{j} + \frac{\partial}{\partial z} (xyz) \hat{k} \\ &= yz \hat{i} + xz \hat{j} + xy \hat{k} \end{aligned}$$

$$\begin{aligned} (\vec{\nabla}\varphi) \cdot \vec{P} &= (yz \hat{i} + xz \hat{j} + xy \hat{k}) \cdot (xy\hat{i} - yz\hat{j} + z^2\hat{k}) \\ &= xy^2z - xyz^2 + xyz^2 = xy^2z. \end{aligned}$$

$$\text{Thus } (\vec{P} \cdot \vec{\nabla})\varphi = (\vec{\nabla}\varphi) \cdot \vec{P} \quad \text{(Showed)}$$

Question: Prove (i) $\vec{v} \cdot (\vec{Q} + \vec{R}) = \vec{v} \cdot \vec{Q} + \vec{v} \cdot \vec{R}$ (ii) $\vec{v} \cdot (\vec{P} + \vec{Q}) = \vec{v} \cdot \vec{P} + \vec{v} \cdot \vec{Q}$ (*yourselvf*)

Solution: Here $\vec{Q} + \vec{R} = y\hat{i} + 2x\hat{j} - 3z\hat{k} + x^2\hat{i} - z^2\hat{j} - y^2\hat{k}$

$$= (x^2 + y)\hat{i} + (2x - z^2)\hat{j} - (3z + y^2)\hat{k}$$

$$\vec{v} \cdot (\vec{Q} + \vec{R}) = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot \{(x^2 + y)\hat{i} + (2x - z^2)\hat{j} - (3z + y^2)\hat{k}\}$$

$$= \frac{\partial}{\partial x}(x^2 + y) + \frac{\partial}{\partial y}(2x - z^2) - \frac{\partial}{\partial z}(3z + y^2) = 2x + 0 - 3 = 2x - 3$$

$$\text{Again, } \vec{v} \cdot \vec{Q} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot (y\hat{i} + 2x\hat{j} - 3z\hat{k})$$

$$= \frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(2x) - \frac{\partial}{\partial z}(3z) = 0 - 0 - 3 = -3$$

$$\text{And } \vec{v} \cdot \vec{R} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot (x^2\hat{i} - z^2\hat{j} - y^2\hat{k})$$

$$= \frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(y^2) = 2x - 0 - 0 = 2x$$

$$\vec{v} \cdot \vec{Q} + \vec{v} \cdot \vec{R} = -3 + 2x = 2x - 3.$$

Thus $\vec{v} \cdot (\vec{Q} + \vec{R}) = \vec{v} \cdot \vec{Q} + \vec{v} \cdot \vec{R}$ (**Proved**)

Question: (a) If $\varphi = x^2yz$ then prove that $\vec{\nabla} \times (\vec{\nabla}\varphi) = 0$.

(b) If $\varphi = xyz$ and $\underline{A} = xz\hat{i} - y^2\hat{j} + z^2\hat{k}$ then show that $\vec{\nabla} \cdot (\vec{\nabla} \times \underline{A}) = 0$.

Solution: (a)

$$\begin{aligned}\vec{\nabla}\varphi &= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \varphi = \frac{\partial\varphi}{\partial x}\hat{i} + \frac{\partial\varphi}{\partial y}\hat{j} + \frac{\partial\varphi}{\partial z}\hat{k} \\ &= \frac{\partial}{\partial x}(x^2yz)\hat{i} + \frac{\partial}{\partial y}(x^2yz)\hat{j} + \frac{\partial}{\partial z}(x^2yz)\hat{k} \\ &= 2xyz\hat{i} + x^2z\hat{j} + x^2y\hat{k}\end{aligned}$$

$$\begin{aligned}\text{Now, } \vec{\nabla} \times (\vec{\nabla}\varphi) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & x^2z & x^2y \end{vmatrix} \\ &= \hat{i} \left\{ \frac{\partial}{\partial y}(x^2y) - \frac{\partial}{\partial z}(x^2z) \right\} - \hat{j} \left\{ \frac{\partial}{\partial x}(x^2y) - \frac{\partial}{\partial z}(2xyz) \right\} + \hat{k} \left\{ \frac{\partial}{\partial x}(x^2z) - \frac{\partial}{\partial y}(2xyz) \right\} \\ &= (x^2 - x^2)\hat{i} - (2xy - 2xy)\hat{j} + (2xz - 2xz)\hat{k} = 0.\end{aligned}$$

$$\text{(b) } \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & -y^2 & z^2 \end{vmatrix}$$

$$\begin{aligned} &= \hat{i} \left(\frac{\partial}{\partial y} z^2 + \frac{\partial}{\partial z} y^2 \right) - \hat{j} \left(\frac{\partial}{\partial x} z^2 - \frac{\partial}{\partial z} xz \right) + \hat{k} \left(-\frac{\partial}{\partial x} y^2 - \frac{\partial}{\partial y} xz \right) \\ &= (0 + 0)\hat{i} - (0 - x \cdot 1)\hat{j} + (-0 - 0)\hat{k} = x\hat{j} \end{aligned}$$

$$\text{Now, } \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x\hat{j}) = \frac{\partial}{\partial y} (x) = 0.$$

Question: If $\varphi = xyz^2$ and $\vec{A} = x^2\hat{i} - z^2\hat{j} - y^2\hat{k}$ then show $\vec{\nabla} \cdot (\varphi\vec{A}) = (\vec{\nabla}\varphi) \cdot \vec{A} + \varphi(\vec{\nabla} \cdot \vec{A})$

Solution: Here $\varphi\vec{A} = xyz^2 \cdot (x^2\hat{i} - z^2\hat{j} - y^2\hat{k}) = x^3yz^2\hat{i} - xyz^4\hat{j} - xy^3z^2\hat{k}$

$$\begin{aligned}\text{So, } \vec{\nabla} \cdot (\varphi\vec{A}) &= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot (x^3yz^2\hat{i} - xyz^4\hat{j} - xy^3z^2\hat{k}) \\ &= \frac{\partial}{\partial x}(x^3yz^2) - \frac{\partial}{\partial y}(xyz^4) - \frac{\partial}{\partial z}(xy^3z^2) \\ &= 3x^2yz^2 - xz^4 - 2xy^3z\end{aligned}$$

$$\begin{aligned}\text{Also, } \vec{\nabla}\varphi &= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot xyz^2 = \frac{\partial}{\partial x}(xyz^2)\hat{i} + \frac{\partial}{\partial y}(xyz^2)\hat{j} + \frac{\partial}{\partial z}(xyz^2)\hat{k} \\ &= yz^2\hat{i} + xz^2\hat{j} + 2xyz\hat{k}\end{aligned}$$

$$\text{Therefore } (\vec{\nabla}\varphi) \cdot \vec{A} = (yz^2\hat{i} + xz^2\hat{j} + 2xyz\hat{k}) \cdot (x^2\hat{i} - z^2\hat{j} - y^2\hat{k}) = x^2yz^2 - xz^4 - 2xy^3z$$

$$\begin{aligned}
\text{Again, } \quad \varphi(\vec{\nabla} \cdot \vec{A}) &= xyz^2 \left[\left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x^2 \hat{i} - z^2 \hat{j} - y^2 \hat{k}) \right] \\
&= xyz^2 \left\{ \frac{\partial}{\partial x} (x^2) - \frac{\partial}{\partial y} (z^2) - \frac{\partial}{\partial z} (y^2) \right\} = 2x^2 y z^2 \\
(\vec{\nabla} \varphi) \cdot \vec{A} + \varphi(\vec{\nabla} \cdot \vec{A}) &= 2x^2 y z^2 + x^2 y z^2 - x z^4 - 2x y^3 z = 3x^2 y z^2 - x z^4 - 2x y^3 z
\end{aligned}$$

$$\text{Hence, } \vec{\nabla} \cdot (\varphi \vec{A}) = (\vec{\nabla} \varphi) \cdot \vec{A} + \varphi(\vec{\nabla} \cdot \vec{A}) \quad \textbf{(Showed)}$$